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Grid Cost and Emissions Externalities of US Data Center Loads with Temoa

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Abstract
Data centers may add gigawatts of load to the US grid in coming years. New generation will be necessary to feed their servers and cooling. The type of generation that grid regulators and industry actors decide to build will impact the nation's grid reliability, electricity costs and emissions. We model possible grid futures exploring 'three pillars' regulation that would require data center loads to be met with renewable energy that is (1) new, (2) hourly-matched, and (3) local. Additional scenarios also explore partial regulation of data centers and uncertainties in future demand, electricity prices, and grid development. These simulations find higher local energy prices and emissions without regulation supporting renewable generation to supply data center loads.

1 Introduction
Data center demands are here, they are growing, they could get pretty big. Cite grid/cost/environmental impacts.

2 Results
Lower LMPs with renewables, less pollution deaths too. higher capex maybe. batteries help out. sensitivity analysis needed in projected demand and costs of fuel and generators.

3 Discussion
Results vary by region and by the sensitivity analysis.

4 Methods
Talk some amount about temoa and enforcing the three pillars. public github. public demo code and maybe notebooks with cute graphics. a little dashboard to explore results.

5 Conclusions
We should build clean energy to feed the servers.

Acknowledgments
Sample text inserted for demonstration.

Funding
Sample text inserted for demonstration.

Author contributions
Sample text inserted for demonstration.

Data availability
Sample text inserted for demonstration.

Supplementary data

Sample text inserted for demonstration.

References